

Research Project: Advanced Visitor Management System

a. Introduction

Visitor management is a critical aspect of organizational security and efficiency. Traditional paper-based systems lead to tracking difficulties, inefficiencies, and waste. This project proposes a digital visitor management system, enhanced with advanced intelligence features to detect suspicious visitors while protecting privacy.

b. Part One: Visitor Management System

Problem:

1. Difficulty in tracking visitors.
2. Excessive paper waste.

Solution:

1. Digital visitor registration (name, national ID, phone number, host, laptop registration if applicable).
2. Visitor pass issuing and return.
3. Visitor time-in and time-out logging.
4. Visitor/host search functionality.

Features:

1. Security/receptionist login.
2. Add visitor information.
3. Search visitor/host feature.
4. Visitor history tracking.
5. Host notifications.

c. Part Two: Advanced Visitor Management System

Problem:

1. Lack of mechanisms to detect suspicious visitors.

Solution:

1. Sharing anonymized security alerts about suspicious visitors.
2. Improving security while protecting visitor privacy.

Reports:

1. Fake identification.
2. Unauthorized access.
3. Theft.

4. Misconduct.
5. Blacklisted visitors.

Shared Intelligence:

1. Incident category.
2. Date.
3. Organization ID.
4. Risk level.

Features:

1. Incident Report: Report incident, select incident type, add description.
2. Intelligent Dashboard: View alerts, risk indicators, number of incidents, reports.
3. Reports: Daily visitors, frequent visitors, incident reports.

4. Technical Architecture

1. Frontend: HTML/CSS, JavaScript, Bootstrap.
 - a. Vanilla-module(esm).
 - b. axios - backend communication.
 - c. sweetAlert2 - attractive alert dialogs.
 - d. bootstrap icons - professional UI.
2. Backend: Django (Python).
3. Database: PostgreSQL.
4. Authentication: Django authentication.
5. Version Control: Git/GitHub.

Data Handling:

1. ID numbers stored raw locally.
2. Encrypted when shared across organizations.
 - a. AES-256 in transit.
 - b. ID hashed with SHA-3.
 - c. TLS 1.3 enforced.
 - d. Zero knowledge model.

5. Existing solutions

1. Vizitor: web application.
 - a. ODPC inspection.
 - b. Kenya DPA 2019.
 - c. NGO and UN campus ready through pre-registration.
 - d. Silicon Savannah digital standards.
2. Soja: web application and mobile application.
 - a. Streamlined and automated registration through ID scan.
 - b. Wireless print based visitor IDs/badges.
 - c. Virtual passes for staff or residents through the use of QR codes.

3. GuestPass: mobile application.

All lacking the incident reporting and a shared but encrypted intelligent network.

6. Expected Outcomes

1. Reduced paper waste and improved visitor tracking.
2. Enhanced security through shared intelligence.
3. Privacy-preserving alerts for suspicious activity.
4. Comprehensive reporting and analytics for management.

7. Conclusion

This project bridges basic visitor management with advanced intelligence features, creating a secure, efficient, and privacy-conscious system. It addresses both operational efficiency and modern security challenges.